Pratik Sharma

Backend Python Developer — AI/ML Engineer

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Professional Summary

Computer Engineering undergraduate with hands-on experience in Python development, Django-based web applications, and machine learning. Familiar with REST API development, relational database design, data preprocessing, and model evaluation. Interested in building reliable backend systems and practical AI/ML applications.

Technical Skills

Languages	Python, C++, C, JavaScript, SQL
Backend Development	Django REST Framework
Databases	PostgreSQL, MySQL
Machine Learning	NumPy, Pandas, Scikit-learn, Matplotlib
Tools	Git, GitHub, Linux, VS Code, Jupyter Notebook
Web Technologies	HTML, CSS
LLM / RAG	RAG, GraphRAG, Neo4j, Embeddings, Vector Search, LangChain

Projects

GrubMate – Online Food Delivery and Real-Time Tracking *Django REST Framework, React*

- Built a food-delivery platform with separate workflows for customers, restaurant owners, and delivery personnel.
- Developed backend API functionality for restaurant and delivery workflows using Django REST Framework.
- Implemented real-time delivery-location updates using WebSockets / Django Channels and displayed tracking on a map with Leaflet.
- Source code: GitHub

E-Commerce Platform **Python, Django, PostgreSQL**

- Developed a full-stack e-commerce application with user authentication, product management, and shopping cart functionality.
- Designed relational database models in PostgreSQL to manage application data.
- Implemented reusable Django backend components and session-based cart functionality.
- Built a responsive user interface for interacting with product and shopping features.
- Source code: GitHub

Machine Learning Classification Project **Python, Pandas, Scikit-learn**

- Cleaned and preprocessed data and performed exploratory data analysis to prepare features for classification.
- Applied feature engineering and feature selection to support model development.
- Trained and evaluated classification models using Scikit-learn.
- Used Matplotlib to visualize data patterns and communicate analytical findings.
- Source code: GitHub

Retrieval-Augmented Generation (RAG) System for Pediatric Diabetes **Python, Embeddings, ChromaDB, LLMs**

- Built a baseline RAG pipeline that chunks and embeds 229 documents on diabetes in children and adolescents using MedCPT.
- Implemented vector similarity search to retrieve relevant passages and generate context-grounded answers with Groq API / GPT-OSS-20B.
- Source code: GitHub

GraphRAG System for Pediatric Diabetes Knowledge Base **Python, Neo4j, LLMs**

- Built a GraphRAG ingestion pipeline for 229 diabetes-related text documents, extracting typed entities and evidence-backed relationships and storing them in Neo4j.
- Designed a Neo4j graph schema and Cypher retrieval that combines vector search with one-hop expansion to provide connected context.
- Integrated retrieved passages and graph facts with Ollama's llama3.1:8b to generate answers grounded in the retrieved evidence, with source references.
- Source code: GitHub

Experience

AI & Machine Learning Intern

Vortex Tech

- Worked on supervised machine learning projects using Python.
- Performed data preprocessing and feature engineering.
- Developed classification models and evaluated model performance.
- Collaborated using Git and Jupyter Notebook.

Education

Bachelor in Computer Engineering 2023 – Present
Purbanchal University Expected Graduation: 2027
Relevant Coursework: Data Structures & Algorithms, DBMS, Machine Learning, Artificial Intelligence, Distributed Systems, Big Data Analytics

Higher Secondary Education (+2 Science) 2021 – 2022
Global School of Science, New Baneshwor, Kathmandu GPA: **3.54**

Secondary Education (SEE) 2014 – 2020
Nexus International School, Pepsicola, Kathmandu GPA: **3.95**